

COUNTING EFFICIENCIES OF SIX COMMERCIAL PARTICLE COUNTERS

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Abstract—Counting efficiencies of a condensation nuclei counter (TSI 3020), a white-light optical particle counter (Climet CI-8060) and four laser instruments (PMS LAS-X, LAS-250X, LPC 525 and HP-LAS) were determined relative to the LAS-X.

Measurements were made in the geometric diameter range of 0.1–4 μm using latex spheres as well as monodisperse organic and inorganic particles produced by a vibrating orifice generator.

The high-pressure in-line counter (HP-LAS) shows a nonlinear response to gas velocity which can be taken into account by a calibration. The lower detection limits (50 % points) of the conventional laser counters (LAS-X and LAS-250X) agree within 0.05 μm with their nominal specifications; for the white-light counter (CI-8060) the actual lower limit is at about 0.4 μm . For all counters, the degree of inlet losses for larger particles varies greatly with inlet design and flow velocity of each counter.

INTRODUCTION

Our present knowledge of the sizing accuracy of light-scattering particle counters (loosely called 'optical particle counters' although they usually do more than just count) is much better developed than our data base of the counting efficiencies of such devices.

There are a number of obvious reasons for this situation. For one, it is possible to generate particles of almost any desirable physical property as calibration aerosols. These particles can easily be sized by independent methods. In addition, Mie theory aided by powerful computer codes permits the calculation of optical equivalent diameters for virtually any combination of scattering angle, wavelength, refractive index, particle size and shape. By contrast it is very difficult to generate good primary standards of particle *concentration*, especially in the operating range of OPCs above 0.1 μm .

There are applications, however, where correct size representation is less important than the ability to measure absolute particle concentrations with good accuracy, either because the particle size in a given experiment is already well defined or, because the nature of the application allows a wider margin of uncertainty for sizes than for concentrations.

The performance of a particle counter with respect to particle concentration is usually expressed in terms of its counting efficiency. The counting efficiency for a properly designed OPC is bounded toward small particles by its gradual inability to detect weak light pulses, and toward large particles by aerodynamic inlet losses.

Gebhart *et al.* (1983) have recently published a comparison of counting efficiencies for 15 commercial particle counters, most of which are white-light instruments with nominal detection limits above 0.3 μm . The authors' own laboratory laser spectrometer served as a reference instrument in this study which included also a commercial laser counter (Royco 226/236) with a detection limit of 0.12 μm . This laser instrument performed well in that it maintained an efficiency of almost 100 % relative to the reference counter between 0.1 μm and about 1 μm , beyond which it showed a decrease in detection efficiency due to inlet losses.

The present study is directed at the counting efficiencies of a group of four laser instruments including the models LAS-X, LAS-250X, LPC 525 and HP-LAS, all manufactured by PMS, Inc., and one white-light counter (Climet 8060). One of these instruments, the LAS-X, is virtually identical in design to the laser counter Royco 226/236 tested by Gebhart *et al.* The LAS-X was thus considered as a secondary reference instrument for the present work.

The initial goal of the study was mainly the determination of inlet losses for OPCs since it was assumed, based on the above mentioned work of Gebhart *et al.*, that laser counters would

have less difficulty in reaching their nominal *lower* detection limits than white-light counters. As it turned out, the laser instruments reached their specified lower detection limits within $0.05\ \mu\text{m}$ or less and thus indeed performed slightly better than the white-light counter, which fell short by about $0.1\text{--}0.15\ \mu\text{m}$.

Further, a condensation nuclei counter Model 3020 of TSI, Inc. was included to evaluate its detection efficiency for larger particles. While there are numerous studies on detection efficiencies of CNCs for ultrafine particles below $0.1\ \mu\text{m}$, the region of sedimentation and inertia dominated losses has never been investigated.

TECHNICAL CHARACTERISTICS OF THE PARTICLE COUNTERS

Table 1 is a summary of particle counter properties regarding sampling flow rate, minimum detectable size and optical system (as far as the light-scattering counters are concerned).

Of the devices listed here, all except one are extractive samplers in the sense that they draw a portion of the aerosol into the instrument for analysis. The inlet systems of these 'conventional' OPCs are shown in Figs 1a and 1b. Their design will be qualitatively useful in subsequent discussions.

The HP-LAS (high pressure laser aerosol spectrometer) is designed to be an *in situ* detector installed as part of a duct or pipeline with diameters between 0.5 in. and 2 in. (*ca* 1–5 cm). A laser beam traverses the detection cell at right angle to the flow as shown in Fig. 2. The scattered light from particles is collected in the near forward direction by two diodes, one for each component of polarized light. The sensing volume is formed within the laser beam by a coincidence measurement that compares the scattering intensities for each component. The sensing volume cross section is quite small with respect to the entire lumen of the cell. The manufacturer quotes a value of $0.25\ \text{mm}^2$; our measurements show a flow-dependent (!) effective cross section of *ca* $0.1\text{--}0.2\ \text{mm}^2$. This unexpected result will be discussed later on.

Due to its design, the volumetric sample flow rate (reduced to standard pressure) of the HP-LAS depends on the axis flow velocity and the pressure in the cell which varies in size with the pipeline diameter. We measured the axis flow velocity in our 1 in. cell as a function of volume flow rate Q at atmospheric pressure p_0 using a miniature hot film flow sensor centered precisely in the cell. Flow rates were obtained by rotameters with accuracies of about 2%. The result of this calibration is shown in Fig. 3 for nitrogen. From these data the following relationship between $v\ (\text{m s}^{-1})$, $Q\ (\text{l min}^{-1})$ and p was derived by linear regression:

$$v = 2.11 + 0.0352 Q \cdot \frac{p_0}{p}. \quad (1)$$

Table 1. Technical characteristics of particle counters

Model	Manufacturer	Sampling flowrate		Nominally smallest detectable size (μm)	Optical system
		Nominal	Actual		
CNC 3020	TSI, Inc.	$0.3\ \text{l min}^{-1}$	$0.307\ \text{l min}^{-1}$	<i>ca</i> 0.01	—
LAS-X	PMS, Inc.	$0.3\ \text{l min}^{-1}$	$0.338\ \text{l min}^{-1}$	0.12	Active cavity laser $35\text{--}120^\circ$
LAS-250X	PMS, Inc.	$3\ \text{l min}^{-1}$	$3.250\ \text{l min}^{-1}$	0.2	Active cavity laser $35\text{--}120^\circ$
LPC 525	PMS, Inc.	$1\ \text{ft}^3\ \text{min}^{-1}$ ($28.5\ \text{l min}^{-1}$)	$1.1\ \text{ft}^3\ \text{min}^{-1}$	0.2	Passive cavity laser $20\text{--}160^\circ$
HP-LAS*	PMS, Inc.	variable†	—	0.3	Classical scatt. laser $6\text{--}40^\circ$
CI-8060	Climet	$1\ \text{ft}^3\ \text{min}^{-1}$ ($28.5\ \text{l min}^{-1}$)	$1.0\ \text{ft}^3\ \text{min}^{-1}$	0.3	White-light $15\text{--}150^\circ$

* Tested with the 1 in. (2.5 cm) cell.

† *In situ* detection at pipeline pressure and flow rate.

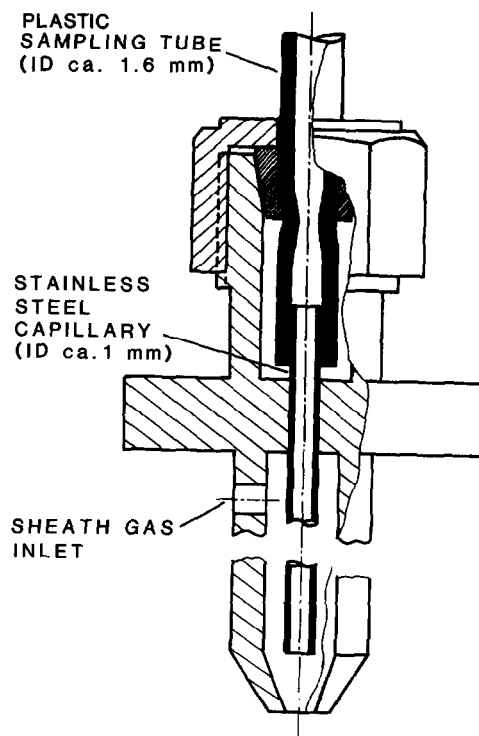


Fig. 1a. Schematic aerosol inlet for LAS-X and LAS-250X.

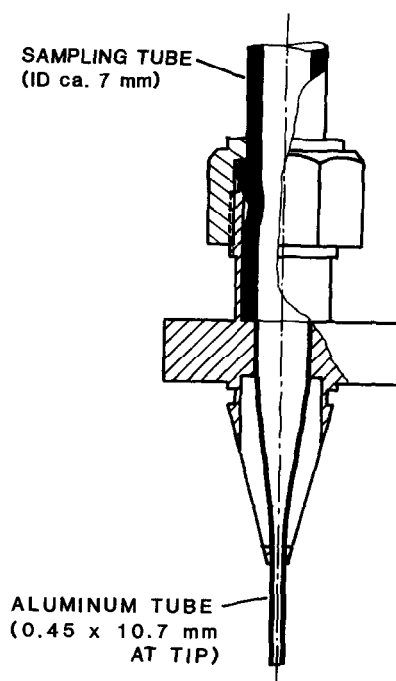


Fig. 1b. Schematic aerosol inlet for LPC-525.

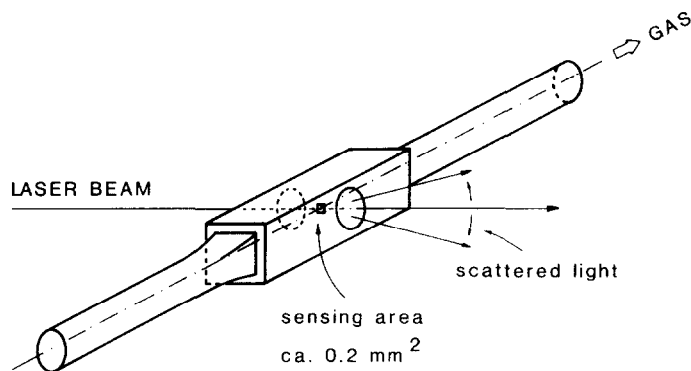


Fig. 2. Schematic diagram of sensing cell for high-pressure in-line counter mod. HP-LAS.

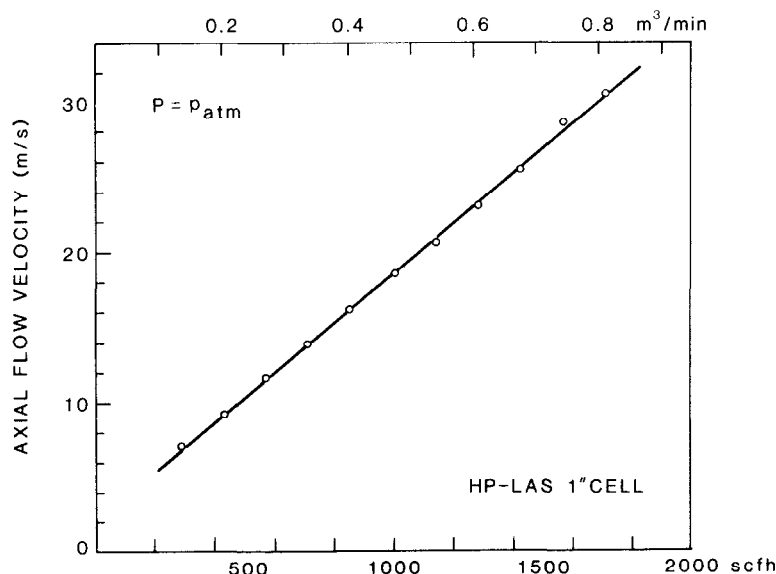


Fig. 3. Gas flow velocity at axis of sensing volume as a function of volumetric flow rate through the pipeline.

EXPERIMENTAL DETAILS

Aerosol properties

Three different types of particles were used, namely latex spheres, and organic and inorganic particles generated by a vibrating orifice generator.

Monodisperse inorganic spheres were produced in the size range of about 0.15–5 μm by dispersing distilled water containing variable amounts of tap water (Fig. 4). The organic particles were generated upward of 0.4 μm by dispersing isopropyl alcohol with additions of a high viscosity organic substance. The organic particles were also spherical but soft and easily deformable. Particle sizes were known very accurately from an independent calibration of the VOG reported elsewhere (Wen *et al.*, 1986).

The latex spheres had nominal diameters of 2.02, 1.09, 0.804, 0.546, 0.481, 0.357, 0.312, 0.261, 0.220 and 0.173 μm . Nominal latex sizes generally agree within 10% with actual sizes in the airborne state (e.g. Kasper and Berner, 1981) and with microscopic measurements, which suffices for the present purpose. It must further be mentioned that latex particles were not used to evaluate the CNC because of the content of ultrafine particles in the aerosol coming from empty droplets.

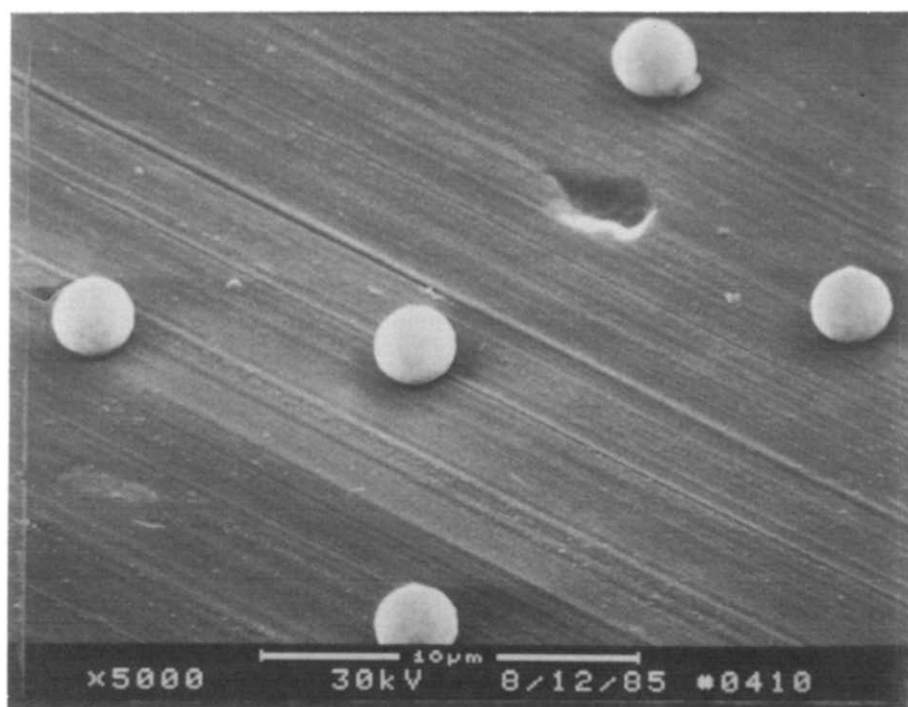


Fig. 4. Scanning electron micrograph of inorganic particles generated with vibrating orifice generator.

In all cases (except for some very specific experiments with the HP-LAS) the particles were in Boltzmann charge equilibrium, i.e. 'neutralized'.

Sampling for extractive counters

In order to avoid unequal sampling losses during any comparison of a given counter with the LAS-X, some consideration was given to the geometry of sampling lines. This was important especially for the higher flow rate instruments since the measurements covered particles up to $4\ \mu\text{m}$. The various configurations are shown schematically in Fig. 5. Cross-sections and flow rates in the various ducts were generally matched to roughly isokinetic conditions. As mentioned before, the aerosol entering the individual ducts was always charge neutralized.

Setup for high-pressure in-line counter

The test setup for the HP-LAS is shown in Fig. 6. The device was installed in a 2.5 cm stainless steel line supplying the necessary flow rates of particle-free gas. Latex particles were injected downstream of the filter; these particles could be neutralized either immediately after the generator (position 1 in Fig. 6) or later in the sampling line leading to the LAS-X

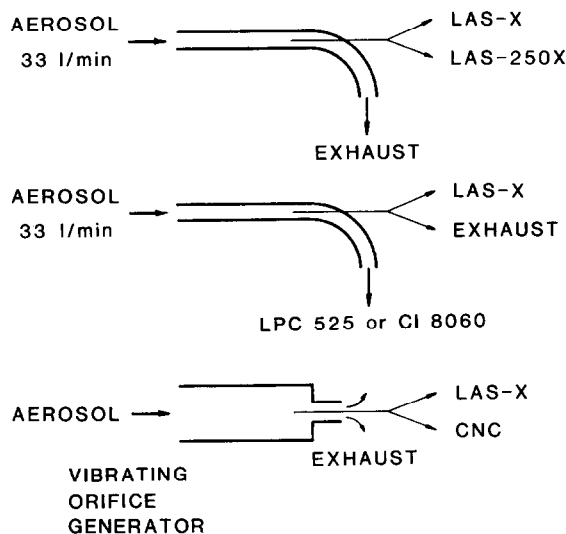


Fig. 5. Sample distribution systems for the instrument comparisons.

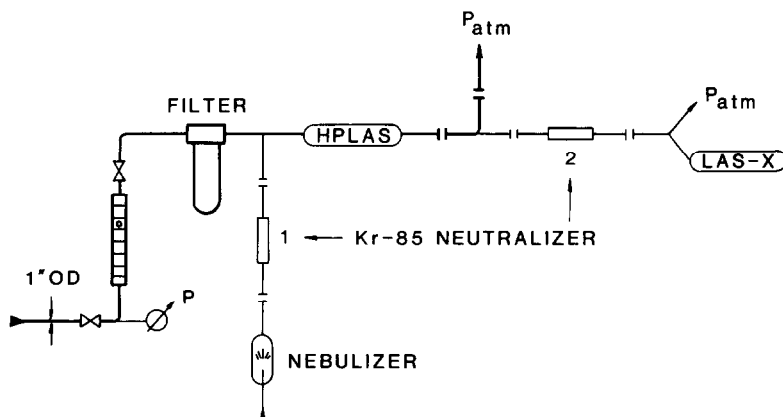


Fig. 6. Experimental setup for calibration of HP-LAS relative to LAS-X.

(position 2). Alternating the neutralizer position or completely removing it from the system permitted us to determine the influence of particle charge on losses in the ducts.

RESULTS AND DISCUSSION

This section presents a summary of the counting efficiency measurements for the five particle counters. All concentrations were measured relative to the Model LAS-X counter.

The LAS-X reference counter

For a better understanding of the subsequent discussion, we shall briefly recall the counting efficiency curve of our reference instrument. In Fig. 7, reproduced from the paper of Gebhart *et al.*, we concern ourselves only with the curves for the Royco 226/236, which is virtually identical in design to the LAS-X used in this study.

The figure shows an efficiency of practically 100% from 0.1 to 1 μm and then a gradual drop to about 70% at 2 μm . This means that subsequent values of counting efficiency relative to the LAS-X for particles smaller than about 1 μm are also correct on an absolute scale; above 1 μm the absolute efficiency is obtained by multiplying the relative efficiency with that of the LAS-X resp. Royco 226/236.

The condensation nuclei counter (CNC)

The CNC Mod. 3020 shows excellent agreement with the LAS-X reference counter from 0.15 to 1 μm (Fig. 8). In this size range it can be considered to count practically all particles present in the gas. Above 1 μm the efficiency curve shows a modest decline relative to the LAS-X. In combination with the declining response of the LAS-X itself (Fig. 7) the true efficiency curve goes through the 50% mark in the vicinity of 3 μm . (One has to keep in mind that losses depend on aerodynamic diameter which may differ from the geometric or optical size.)

The count rates shown in Fig. 8 as well as all subsequent data are corrected for the actual sampling flow rates of the instruments (cf. Table 1).

The measurements reported here cover the CNCs response to large particles. The size range below 0.1 μm down to about 0.005 μm has in the past been the focus of several studies,

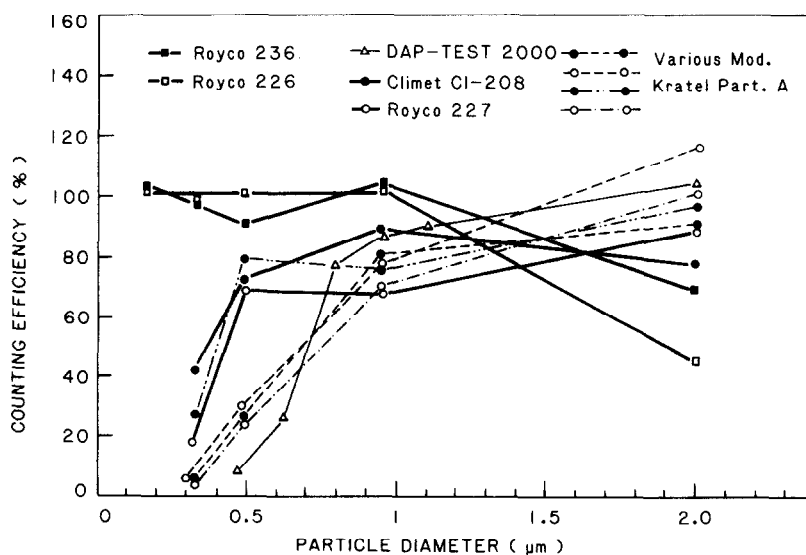


Fig. 7. Absolute counting efficiency of various optical particle counters, including Royco 226/236. (From Gebhart *et al.*, 1983.)

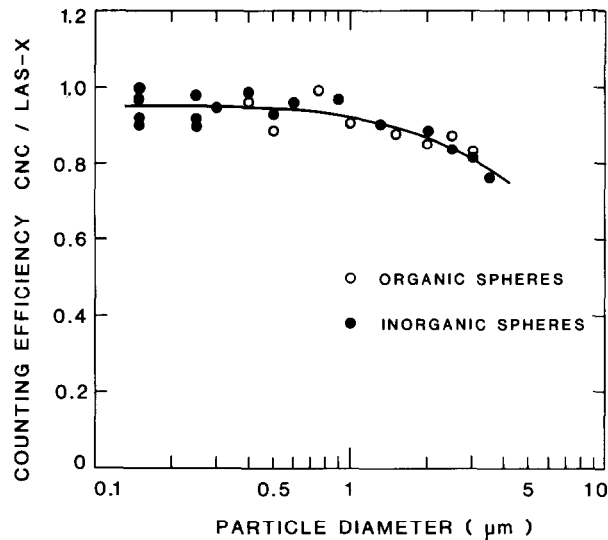


Fig. 8. Counting efficiency curve of condensation nuclei counter 3020 relative to LAS-X.

for this and other types of CNC (e.g. Aggarwal and Sem (1980), for the TSI Instrument). By combining these data one can cover the entire response range of the TSI CNC. This has been done in Fig. 9, using some unpublished data of Niida, Reischl and Kasper (1982).

The optical particle counters

The counting efficiency curves for LAS-250X, LPC 525 and CI-8060 relative to the LAS-X are shown in Fig. 10. Since the LAS-X can detect smaller particles than the other three, the curves in Fig. 10 show not only the performance of these devices for large particles, where aerodynamic inlet effects are dominant, but also the decline in counting efficiencies toward their lower nominal detection limits.

Aerodynamic inlet losses can, for the present purposes, be characterized as parasitic processes which prevent particles from reaching the sensing volume of a counter. They are

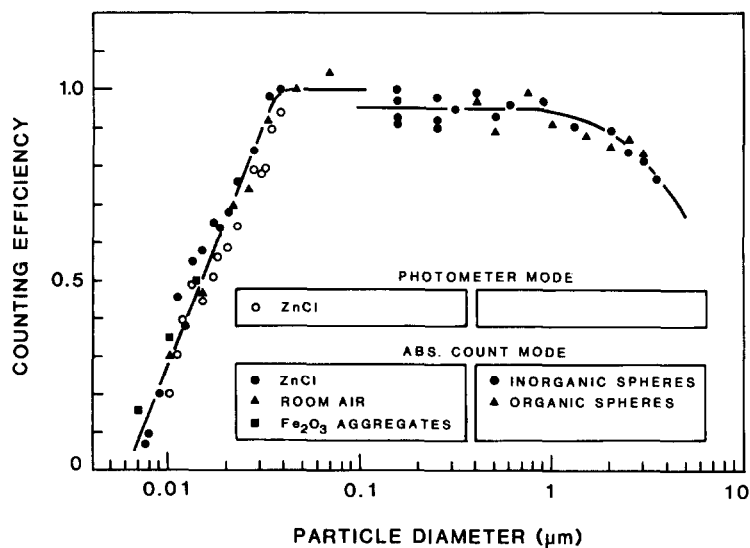


Fig. 9. Counting efficiency of CNC 3020 combining the data of Fig. 8 and unpublished measurements by Niida, Reischl and Kasper (1982) for the size range below 0.1 μm .

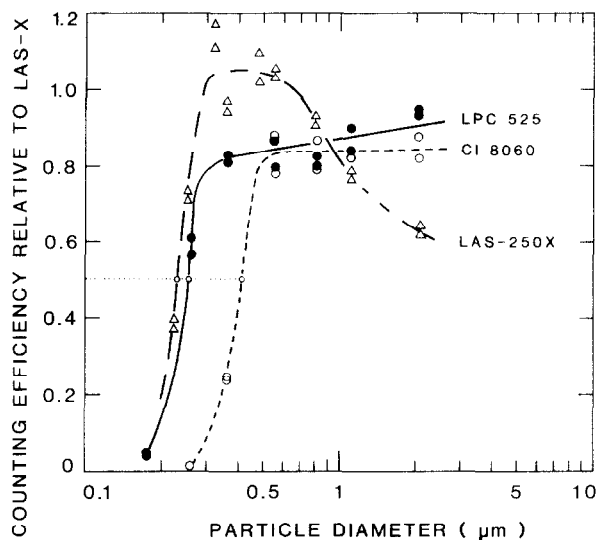


Fig. 10. Counting efficiency curves of LPC 525, CI-8060 and LAS-250X relative to LAS-X.

usually caused by sedimentation and inertial deposition in the ducts and nozzles leading to the sensing zone. Thus, aerodynamic losses generally increase with particle size (exceptions will be noted below). They are also a strong function of inlet geometry and flow field and, hence, very much dependent on the specific design of a device.

As opposed to the aerodynamic and flow related causes for decline in counting efficiency toward the large-particle end of the measuring range, the decrease at the low end of the detection range is due to phenomena in the optical and electronic sensing system of a counter that are related to the sensitivity for weak light pulses near the signal threshold. Among the fundamental causes for reduced detection probabilities are the fact that light pulses are superimposed on the electronic noise, and thus may or may not surpass the detection threshold. Variations of light intensity across the sensing volume are similarly pivotal for weak pulses. The maintenance condition of an optical counter enters as a cause of decreased counting efficiencies through aging effects of the light source as well as misalignment between light beam and particle beam; the latter especially can broadly affect the response of an optical counter (Helsper and Fissan, 1980). Furthermore, since the (adjustable) detection threshold of a counter is usually close to the noise level, there may be a trade-off between its internal noise level and its lower practical size limit. In case of equipment manufactured for use in ultra clean gas analysis, this balance may be tipped in favor of a low internal noise level.

The LAS-250X (Table 2) has a lower limit of $0.2 \mu\text{m}$ according to the manufacturer. Figure 10 shows an efficiency of about 40% for the latex diameter $0.220 \mu\text{m}$, which is quite close considering the uncertainty inherent in most latex sizes. With the next size of $0.261 \mu\text{m}$ the 100% level is reached. The instrument had undergone careful alignment and adjustment of its gain.

The LPC 525 (Table 2) is also a $0.2 \mu\text{m}$ instrument. It evidently does not reach the 100% level anywhere, but it counts 80–95% of the particles above $0.35 \mu\text{m}$. The 50% point of the efficiency curve according to Fig. 10 is near 0.24 or $0.25 \mu\text{m}$. The instrument was new.

The CI-8060 (Table 2) is a $0.3 \mu\text{m}$ instrument using a white-light source. Above *ca* $0.5 \mu\text{m}$ it counts 80% or more of the incoming particles without ever reaching 100%. At $0.36 \mu\text{m}$ the efficiency is only about 25%. The 50% point thus probably lies between 0.40 and $0.45 \mu\text{m}$. The instrument was new.

The above data present evidence that the 50% detection probability points of the laser counters agree within a few hundredths of a micrometer with manufacturer specifications when the instruments are in top condition. It should be noted, however, that misalignments (common transportation problems) or misadjusted amplifier gains can have a noticeable effect on the lower detection limit (Fig. 11). The uppermost curve in Fig. 11 is that for a

Table 2. Counting efficiencies of three optical particle counters relative to LAS-X

Particle size (μm)	LAS-250X		LPC 525		CI 8060	
	(%)	(cm^{-3})	(%)	(cm^{-3})	(%)	(cm^{-3})
2.020	64.0 62.6	4.0	94.4 93.1	0.5	87.8 81.9	2.0
1.091	79.2 76.5	8.2	83.9 90.1	0.7	83.1 81.9	2.0
0.804	93.4 90.8	3.8	82.3 79.8	1.5	78.8 86.3	1.3
0.546	103.7 105.6	16.3	86.8 79.4	2.2	78.3 88.5	2.3
0.481	101.8 109.5	1.7	—	—	—	—
0.357	93.6 96.7	2.4	81.0 83.4	1.9	23.4 24.0	1.7
0.312	109.0 118.0	1	—	—	—	—
0.261	72 74	25	60.6 56.2	2.1	0.5 0.5	2.8
0.220	37 39 39	1.8	— — —	—	— — —	—
0.173	— —	—	4.1 4.6	2.3	— —	—

The left column is the respective counting efficiency relative to LAS-X in % corrected for actual flow rate; the right column is the respective particle concentration at which the ratio was obtained.

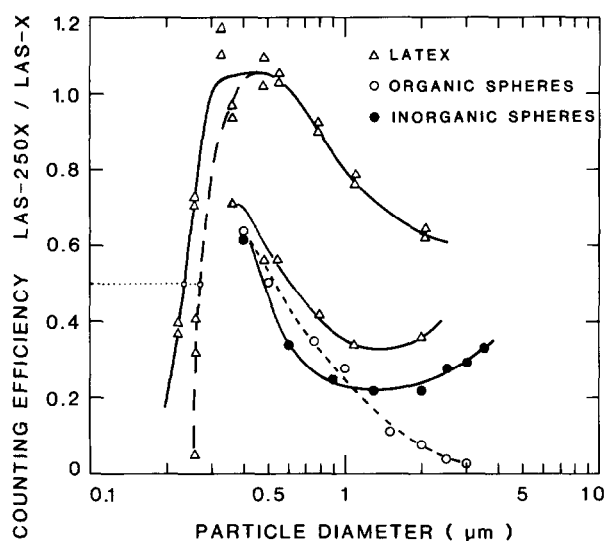


Fig. 11. Counting efficiency curves of LAS-250X under irregular operating conditions. Dashed curve to the left represents drop in counting efficiency due to misalignment or misadjusted gain. Three curves at lower right show counting efficiency as a function of varying degree of inlet losses. (Particles have different aerodynamic diameters; some of them (open circles) are of a viscous oily nature which prevents them from rebounding after deposition.)

properly performing LAS-250X. The dashed curve just below indicates the shift in detection efficiency for a gain offset or misalignment amounting to a particle size shift of just under one size channel.

Toward larger particles, the LPC-525 and the CI-8050 show no noticeable difference in relative counting efficiency up to *ca* 2 μm . Inlet losses do not seem to play a significant role; however, one has to keep in mind that the reference instrument starts to drop above *ca* 1 μm .

The LAS-250X persistently shows a remarkable efficiency drop toward larger particles beginning at *ca* 0.7 or 0.8 μm . During an investigation of the cause for this early drop, the observation was made that the losses were related to the way in which the sampling tube was connected to the aerosol inlet of the counter: the losses could be increased with the degree of tightening of a fitting that holds this tube in place (Fig. 1a). Figure 11 shows a comparison of two series of measurements with the LAS-250X, one with a rigidly tight inlet tube (lower group of three curves), the other with a loose fitting (upper curve) representing the best possible conditions. (The latter data are the ones included in Fig. 10.) When the sampling tube is squeezed by the fitting, the soft tube is necked down to form a venturi nozzle behind which a significant amount of deposition takes place. These deposits are actually visible on the tube walls. Even after correction for this problem, however, the response curve of the LAS-250X did not return to the levels that would be expected from this type of device.

An interesting detail to be noted on Fig. 11 is the upturn of two of the lower curves, namely the ones for latex and for inorganic salt particles. By contrast, the curve for soft organic spheres continues to decrease toward large particles. It appears that the larger elastic and hard particles have enough kinetic energy to bounce back into the gas stream after hitting the walls of the inlet systems.

Most of the inlet losses of the LAS-250X (under proper operating conditions) seem to occur at the transition from the plastic inlet tube to the stainless steel capillary tube (Fig. 1a) where a sharp step occurs in the aerosol flow. Although the geometric dimensions of this particular point are almost identical for the LAS-X and LAS-250X, the losses are far more severe for the latter due to its ten fold sampling rate.

Calculations of impaction losses at the transition were made for both LAS-X and LAS-250X using a model proposed by Pich (1964) which is useful for such estimates. Assuming mean flow velocities of 2.5 and 25 m s^{-1} , respectively, and geometries that were measured directly on the instruments, the following results were obtained.

Impaction losses for the LAS-X increase from 4 to 27 % for aerodynamic diameters from 1 to 3 μm . This is in reasonable agreement with the data of Gebhart *et al.* (29 % for 2.02 μm latex particles).

Absolute losses for the LAS-250X increase from 22.5 to 53 % in the same size interval of 1–3 μm . These values, translated into relative losses, account for most of the decrease in efficiency observed in Figs 10 and 11 for the LAS-250X.

The HP-LAS

The HP-LAS is different from the other counters in that inlet losses are immaterial as long as the device is properly installed. The system was thus only evaluated with two sizes of latex particles. The measurements in Figs 12a and 12b are represented somewhat differently from the previous data in that they show sensing volume cross-section A vs axis flow velocity v . A is considered the important quantity which together with v and p determines the sampling flow. A is determined by the expression

$$A = \frac{N}{c \cdot v \cdot t}, \quad (2)$$

where N = number of particles counted by the HP-LAS

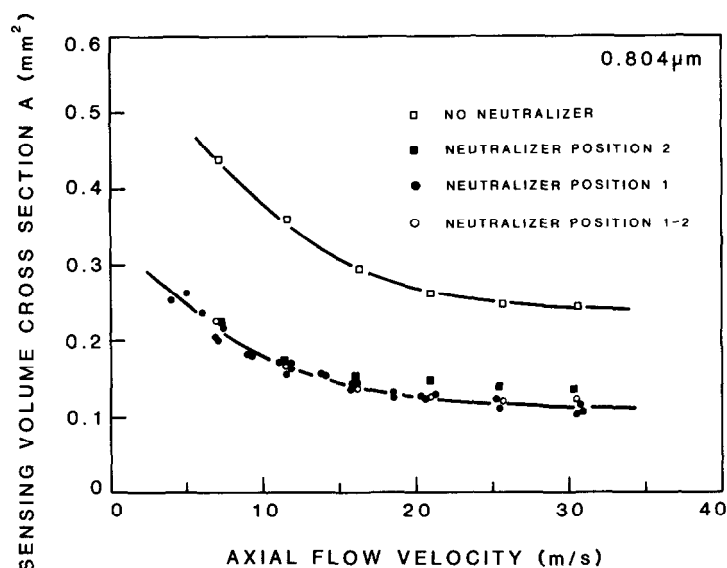
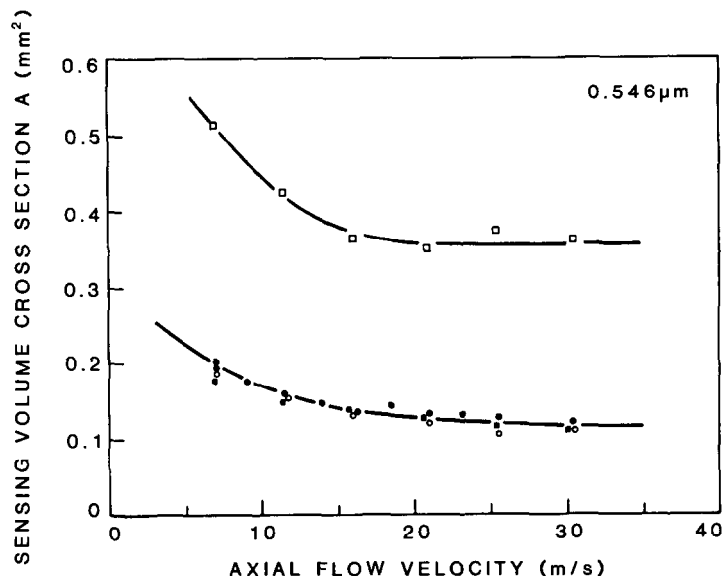
c = particle concentration from LAS-X

t = sampling time

v = axis flow velocity (compare equation (1)).

A is a small portion of the total cross-section, namely the region near the axis of the tube where particles are detected and counted. Since A is defined optically in the HP-LAS one could expect it to be a constant.

An important and surprising conclusion from Fig. 12 is the fact that the 'geometric' cross-sectional area of the sampling volume is a non-linear function of the flow velocity. It decreases systematically from about 0.2 mm^2 at low flow velocities to about 0.1 mm^2 at 20 m s^{-1} ; above 20 m s^{-1} the curves turn flat. In this regime the HP-LAS also begins to shift the

Fig. 12a. Results of HP-LAS calibration at 0.8 μm .Fig. 12b. Results of HP-LAS calibration at 0.5 μm .

particle size distribution toward smaller particles (Fig. 13). Representatives of the manufacturer explain that the pulse processing electronics of the counter are too slow above $ca\ 20\text{ m s}^{-1}$ and systematically interpret light pulses as coming from smaller particles. Consequently, the gas velocity of 20 m s^{-1} represents an upper limit for the operation of the device. However, this limit is not very sharp and, as Fig. 13 indicates, some distortions of the size distribution are already visible at 16 m s^{-1} . The continuous decline in counting efficiency of the HP-LAS even at low gas velocities is currently very difficult to explain.

A second, important conclusion from these data is the fact that the non-linear response is not dependent on particle size in the region of greatest interest to high purity gas applications. Figure 14, where the data of Figs 12a and 12b are superimposed, shows no significant difference between 0.5 and $0.8\ \mu\text{m}$ particles within the limits of accuracy of the calibration. (The region above 20 m s^{-1} is ignored.)

The third important conclusion is that electrical neutralization of the PSL particles is very critical for losses in the sampling line. However, the data show that it does not matter whether

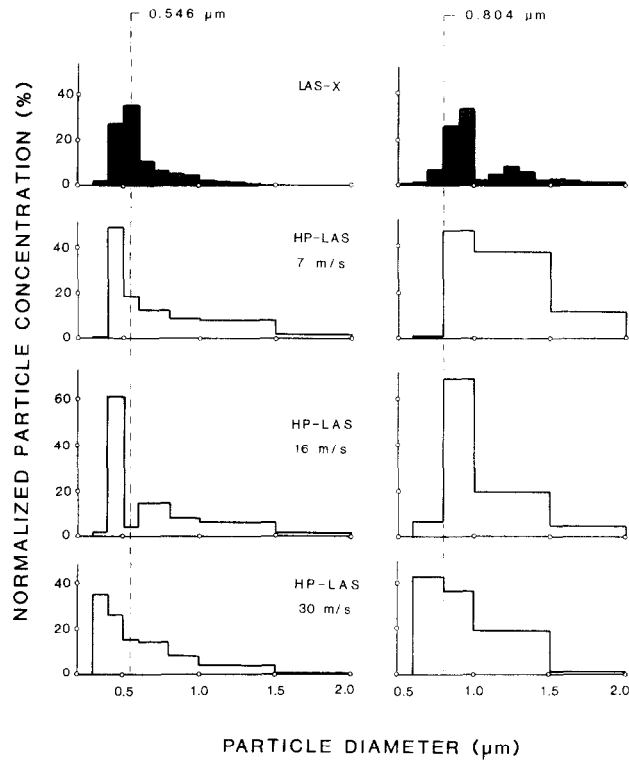


Fig. 13. Normalized size distributions for 0.804 and 0.546 μm PSL spheres. Compared with the LAS-X the HP-LAS shows a shift toward smaller sizes starting at about 16 m s^{-1} .

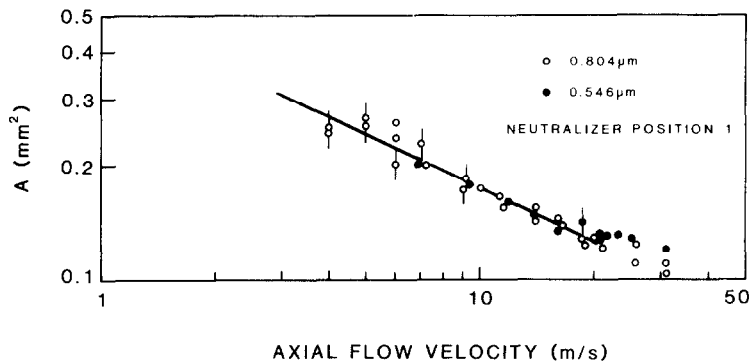


Fig. 14. Superimposed response curves for 0.5 and 0.8 μm (from Figs 12a and 12b).

particles are neutralized immediately after the aerosol generator or after the sampling point to the LAS-X. In other words, it is not the HP-LAS which is charge sensitive but the flow system leading into the LAS-X.

For practical purposes, the non-linear function $A(v)$ can be approximated from about 3 to 20 m s^{-1} by a power law:

$$A = 0.57 v^{-0.47}. \quad (3)$$

Finally, in Fig. 15 the actual sampling flow rate of the HP-LAS with a 1 in. (2.5 cm) cell is plotted as a function of pipeline flow rate and pressure.

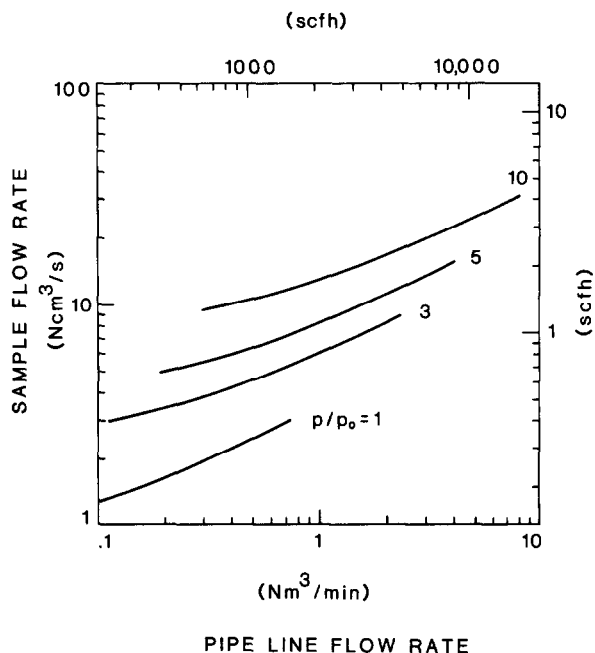


Fig. 15. Sampling flow rate of HP-LAS (1 in. cell) as a function of pipe line flow rate and pressure p/p_0 .

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